

Bipin Kumar Chaudhary

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SUMMARY

Full Stack Engineer with 3+ years of experience building scalable web systems across video streaming, enterprise analytics, and cloud-native applications. Improved video startup latency by **40%** using HLS-based adaptive streaming, reduced database query latency by **30%** via SAP HANA optimization, and enhanced UI performance by **20%** through Angular rendering strategies. Strong in system design, distributed systems, and AI-powered features. **300+** **LeetCode** problems solved.

EXPERIENCE

- Independent Full-Stack Developer (Self-Employed)** | Remote Oct 2023 - Present
- Built and deployed production-grade applications including a video streaming platform and a clinic management system with real-world users.
 - Designed end-to-end system architecture covering authentication, scheduling engines, caching strategies, and asynchronous workflows.
 - Integrated third-party services (WhatsApp API, OpenAI) to enable real-time communication and AI-powered features.
- SAP Labs** | Graduate Research Engineer | *iXp Internship, M.Tech MNNIT Allahabad* Aug 2024 - Jun 2025
- Performed functional and regression testing for SAP Business Application Studio (BAS), ensuring reliability of development workflows and platform features.
 - Developed automated test scripts to convert manual test cases into repeatable validation pipelines, improving testing efficiency.
 - Validated database artifacts using SAP HANA Database Explorer and tested features like the HANA Vector Engine for correctness and stability.
- Nagarro** | Full Stack Developer May 2021 - Sep 2023
- Improved frontend performance by **20% TBT reduction** using Angular OnPush strategy and reusable component architecture, maintaining **60 FPS** under high data load.
 - Designed and developed scalable backend services for a rail asset management system managing **170K+ assets**, enabling sub-second real-time updates.
 - Enhanced engineering productivity by standardizing CI/CD pipelines and unit testing (Jasmine), improving deployment speed by **15%** and mentoring junior developers.

PROJECTS

- StreamSphere** | *Angular, Node.js, AWS, MongoDB, Redis* [Live] | [GitHub]
- Architected a **serverless video processing pipeline** (S3 → Lambda → FFmpeg → HLS) generating adaptive bitrate streams (360p/720p), reducing playback startup time by **40%**.
 - Implemented **event-driven caching** with Redis and Lambda webhooks, reducing feed latency by **30-40%** and S3 origin load by **25%**.
 - Designed **cursor-based pagination** for infinite scroll, ensuring O(1) query performance and consistency under concurrent uploads.
 - Built a fault-tolerant **AI enrichment pipeline** (Whisper, GPT, HuggingFace) using async orchestration, allowing graceful degradation without blocking core workflows.
- DentalConnect** | *Next.js, PostgreSQL (Supabase), Vercel, WhatsApp API* [Live] | [GitHub]
- Designed a full-stack **appointment management platform** with OTP-based authentication via WhatsApp and role-based access (doctor/patient) using JWT in httpOnly cookies.
 - Built a **dynamic scheduling engine** that computes real-time availability by reconciling weekly schedules, blocked dates, partial slot blocks, and existing bookings.
 - Implemented **asynchronous notification workflows** (OTP, confirmations, reminders, cancellations) using WhatsApp Business API and **Vercel Cron** for scheduled automation.
 - Enforced strong data security using **Row-Level Security (RLS)**, input validation (Zod), OTP rate limiting, and middleware-based route protection.
 - Developed a **doctor dashboard with live queue management**, appointment lifecycle tracking, review moderation, and operational analytics.
 - Built **API observability infrastructure** with request logging, latency tracking, and severity classification, along with automated log retention via cron jobs.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, C++, Java, SQL
Core Concepts: Data Structures & Algorithms, System Design, OOP, Concurrency, Caching, Distributed System, REST APIs
Frontend: Angular, React, Next.js, HTML5/CSS3, Tailwind CSS
Backend: Node.js, Express, REST APIs, CI/CD, Docker, Git
AI/ML: OpenAI APIs, Whisper (ASR), HuggingFace Transformers, RAG, Vector Embeddings
Cloud & Databases: AWS (S3, Lambda, CloudFront, EventBridge), SAP BTP, SAP HANA, MongoDB, Redis, PostgreSQL

EDUCATION

Motilal Nehru National Institute of Technology Allahabad | M.Tech(CSE) | **CPI: 8.75/10**
Includes SAP Labs iXp internship (Aug 2024 - Jun 2025) Aug 2023 - Aug 2025
Bundelkhand Institute of Engineering and Technology | B.Tech(CSE) | **78.6%** Aug 2017 - Aug 2021